

Rework Proposal & Experimental Plan

Policy-Hard, Value-Easy: Inverted Actor-Critic Asymmetry in Deep RL for Assistive Indoor Navigation

Working framing: **Architectural Ablations for Sparse-Reward Navigation**
(narrative locks in after Phase 1 multi-seed replication)

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Team size: 2 (1 lead + 1 collaborator)

Compute: 1 cloud GPU (Colab Pro tier)

Status: post-rejection rework (DLI 2026, 3 reviewers)

1. Executive Summary

This document is the working plan for reworking the rejected DLI 2026 submission into a substantially stronger paper. It synthesises the three reviewers' feedback, fixes the methodological gaps they identified, and adds the experimental rigour standard for RL venues in 2026.

The starting point is honest: the reviewers were correct on every major point. The core idea (inverted actor-critic asymmetry) is interesting and worth defending, but the submitted version did not actually demonstrate it. Single-seed results, no statistical reporting, one fixed map, an Exp 3 configuration that contradicts the hypothesis, a POMDP framing inconsistent with the state vector, and dollar-store baselines all combined to make the central claim unfalsifiable from the evidence presented.

What this rework will change

- Multiple seeds (5–10 per configuration) with stratified-bootstrap 95% confidence intervals, IQM aggregation, and pairwise significance tests via the rliable framework. This is non-negotiable and is Phase 1.
- A clean capacity-ratio ablation that holds total parameter budget constant, isolating the actor:critic ratio as the independent variable. This is what saves (or kills) the inverted asymmetry hypothesis.
- Procedurally generated map layouts with held-out test sets, so generalisation claims are supported by evidence rather than assertion.
- A fixed POMDP framing: either drop normalised global coordinates from the state vector (preferred) or relabel as a fully-observed MDP. No more handwaving.
- New evaluation metrics: path optimality ratio against A* shortest paths, near-miss frequency, wait-action usage, and per-room success rates.
- Cross-environment validation on 2–3 standard sparse-reward benchmarks (MiniGrid FourRooms, MultiRoom, DoorKey). The hypothesis is a general claim and must be tested on more than one map.
- Stronger baselines: Double DQN, Dueling DQN, and A2C in addition to vanilla DQN and PPO.
- Empirical Lipschitz estimates for π and V , replacing the current qualitative argument with measurement.
- An implemented and evaluated PPO+DQN hybrid safety architecture, addressing the reviewer comment that the proposal was speculative.
- Optional Phase 4 stretch goals: GRPO comparison, sensor-noise robustness ablation.

What this rework deliberately will NOT do

- **Continuous action spaces.** Out of scope. Switching from discrete to continuous changes the algorithms (DQN drops out, SAC/TD3 enter), doubles the experimental matrix, and addresses a different research question (action-space design) than the central one (capacity asymmetry). It is mentioned as future work in the paper.
- **Full sim-to-real deployment.** Hardware deployment is mentioned as future work; we add sensor-noise ablations to address robustness concerns without leaving simulation.
- **Vision-Language-Model integration.** Crowded space dominated by larger labs; no clear angle for this paper. Stays as a future work bullet.

Decision philosophy

The narrative framing for the paper is left provisional. The default working frame is "architectural ablations for sparse-reward navigation," with inverted asymmetry as one finding among several. After Phase 1 (multi-seed replication of the original Exp 4 configuration), the evidence will determine whether asymmetry survives as the headline claim or recedes to a supporting finding. This is the right epistemic posture: let the data choose the story.

2. Reviewer Feedback Synthesis

The three reviewers' critiques cluster into seven concrete problems. Each is mapped below to the planned response.

ID	Reviewer concern	Planned response
R-1	No multiple seeds, no error bars, no significance tests. Differences between Exp 1 (41.0), Exp 3 (40.9), and Exp 4 (41.1) are likely within RL noise. (All 3 reviewers.)	Phase 1: Re-run every Table 1/2 configuration with 10 seeds. Report IQM $\pm$ 95% stratified-bootstrap CI (reliable). Pairwise Mann–Whitney U for ordinal comparisons. Performance profiles for distribution.
R-2	Exp 3 (small actor, deep critic) achieves $\approx$ Exp 4 performance, directly contradicting the inverted asymmetry hypothesis. Is the effect parameter count rather than ratio? (Reviewer 3.)	Phase 2: Capacity-ratio ablation. Fix total parameter budget at 5 levels (e.g., 100k, 200k, 400k, 800k, 1.6M). Sweep actor:critic ratios {1:4, 1:2, 1:1, 2:1, 4:1} at each. Decouples ratio from total count. Three possible outcomes plotted explicitly.
R-3	POMDP framing inconsistent with state: normalised (x,y) coordinates effectively give global pose. (Reviewer 1.)	Phase 2: Replace global coords with relative target vector (Δx , Δy normalised by max distance) or remove entirely. Test both. If we keep partial observability, add LSTM/GRU encoder ablation to justify POMDP claim.
R-4	Single fixed map. Generalisation and deployment claims unsupported. (All 3 reviewers.)	Phase 2: Procedural layout generator. Train on $N_{\text{train}} \in \{10, 50, 200\}$ held-out maps, evaluate on 50 unseen test maps. Report train-test generalisation gap. This is what converts "deployment potential" from claim to measurement.
R-5	Sample efficiency comparison in episodes is misleading; should normalise by environment steps. (Reviewer 1.) Figure 3 says ~ 100 episodes but text says ~ 20 . (Reviewer 1.)	All Phase 1+ reporting in environment steps, not episodes. Sample efficiency = first env step reaching 90% of asymptotic IQM. Figure inconsistencies are resolved by full retrain.
R-6	Weak baselines (only vanilla DQN, PPO). Need Double/Dueling DQN, A2C, etc. (Reviewer 1.) No non-RL baseline. (Reviewer 3.)	Phase 1 baseline set: vanilla DQN, Double DQN, Dueling DQN, A2C, PPO (symmetric), PPO (inverted-asymmetry). Phase 1 also adds A* shortest-path as non-RL optimality reference; success-rate-and-step counts contextualised against A*.
R-7	Lipschitz argument qualitative only; needs measurement. (Reviewer 1.)	Phase 3: Empirical local Lipschitz estimates. Sample state pairs, compute $\ \pi(s) - \pi(s')\ _{\infty} / \ s - s'\ _1$ and same for V. Plot distributions. Compare across configurations.
R-8	Hybrid PPO+DQN safety architecture proposed	Phase 4 (optional but high value):

	but not tested. "Test it or label it speculation." (Reviewer 2.)	Implement PPO planner + DQN safety filter. Compare hybrid vs. PPO-alone vs. DQN-alone on collision rate, steps-to-goal, success rate. Promotable to a separate paper if results are strong.
R-9	DQN's 100% success rate might be trivial: does it succeed by camping (Wait action)? (Reviewer 3.) Wait usage never reported. (Reviewer 3.)	All Phase 1+ runs log per-action frequency. Wait-action share reported. If DQN's success is dominated by Wait, the "safety" claim is degenerate and we report it as such.
R-10	Path traversal steps reported without optimum baseline; "14.3 steps" is uninterpretable. (Reviewer 3.)	Compute A*-shortest-path length per episode. Report Path Optimality Ratio = $\text{agent_steps} / \text{A*_steps}$. Optimal policy $\rightarrow 1.0$.
R-11	Figure 2–5 illegible; LaTeX bug in §3.3 ("265" coefficient); Tsai citation year wrong. (Reviewer 3.)	Figure regeneration in Phase 4 with seaborn defaults, larger fonts, cleaned x-axis. LaTeX cleanup pass before resubmission.
R-12	Static-only environment; no dynamic obstacles. (Reviewer 2, Reviewer 3.)	Phase 3 ablation: dynamic-obstacle variant (1–2 moving NPCs on simple patrol paths). Used as ablation, not main training setting, to avoid confounding the asymmetry result.

Two cosmetic reviewer comments are deferred to the writing phase: simplifying the abstract (Reviewer 2) and moving the colour-coding description to an appendix (Reviewer 3).

3. Research Questions and Hypotheses

The submitted paper conflated several distinct claims into one narrative. We separate them so each can be falsified independently.

3.1 Research questions

- **RQ1.** In sparse-reward grid-world navigation, does inverted actor-critic capacity asymmetry (deeper π , shallower V) outperform symmetric and conventional configurations when total parameter budget is held constant?
- **RQ2.** Does this advantage generalise across grid layouts, or is it specific to the original 20×20 residential map?
- **RQ3.** Does the empirical local Lipschitz behaviour of learned π and V match the theoretical prediction that $\text{Lip}(\pi) \gg \text{Lip}(V)$?
- **RQ4.** Does the hypothesis transfer to standard sparse-reward benchmarks (MiniGrid family)?
- **RQ5.** Can a PPO-planner + DQN-safety-filter hybrid deliver PPO's path efficiency and DQN's zero-collision safety simultaneously?

3.2 Pre-registered hypotheses

Pre-registering hypotheses before running experiments protects against the post-hoc narrative drift that made the original paper unfalsifiable.

ID	Hypothesis	Falsification criterion
H1	At fixed parameter budget, inverted asymmetry (actor:critic > 1) achieves higher IQM return than symmetric (1:1) and conventional (< 1) ratios.	FALSE if symmetric or conventional matches or beats inverted within overlapping 95% CIs at all 5 budget levels.
H2	The advantage is attributable to the ratio, not the total count: at fixed ratio, returns scale modestly with budget but the ratio's effect dominates.	FALSE if ratio-effect-size < budget-effect-size in a two-way ANOVA, or if the ratio coefficient's CI crosses zero.
H3	Inverted asymmetry generalises: train/test gap on procedural layouts is smaller for inverted than for conventional configurations.	FALSE if generalisation gap is statistically indistinguishable across ratios (overlapping CIs).
H4	Empirical local Lipschitz estimate of learned policies exceeds that of learned value functions by $\geq 2\times$ across configurations.	FALSE if the Lipschitz ratio $\text{Lip}(\pi)/\text{Lip}(V) < 1.5$ or has 95% CI crossing 1.0.
H5	On MiniGrid sparse-reward tasks, inverted asymmetry produces measurable sample-efficiency gains over symmetric PPO.	FALSE if no significant improvement on ≥ 2 of the 3 MiniGrid tasks (Mann–Whitney $p < 0.05$).
H6	Hybrid PPO+DQN achieves PPO's average step count within 10% AND DQN's collision rate within 1 percentage point.	FALSE if either constraint is violated.

Each hypothesis has a clear pre-committed test. Failure is informative, not a project failure.

4. Methodological Upgrades

4.1 Evaluation protocol

We adopt the reliable framework (Agarwal et al., NeurIPS 2021, Outstanding Paper) as the project-wide evaluation standard.

- **Aggregate metric:** Interquartile Mean (IQM) over seeds and tasks, computed after dropping the bottom and top quartiles. Robust to single-seed outliers in a way that mean is not.
- **Uncertainty:** 95% stratified-bootstrap confidence intervals via percentile method, with $B = 2000$ resamples. No assumption of normality.
- **Pairwise comparison:** Probability of improvement (reliable) and Mann–Whitney U for ordinal cases. Holm correction for multiple comparisons within an experiment family.
- **Performance profile:** Fraction of runs achieving each score threshold; reveals distribution shape that means hide.
- **Seeds:** 10 per configuration for main results; 5 for exploratory sweeps; 3 for sanity checks only.
- **X-axis:** environment steps, never episodes. Episode-based comparisons across algorithms with different rollout sizes are misleading.
- **Convergence definition:** first env step at which the 10-seed IQM crosses 90% of its asymptotic value.

4.2 New evaluation metrics

These metrics directly respond to reviewer asks and substantially strengthen behavioural analysis.

Metric	Definition and rationale
Path Optimality Ratio (POR)	$\text{agent_steps} / A^*_{\text{shortest_path_steps}}$, per episode. Optimal = 1.0. Contextualises raw step counts against the geometric minimum. Reviewer 3 explicitly asked for this.
Success Rate (SR)	Fraction of evaluation episodes reaching the target room. Reported with binomial 95% CI.
Collision Rate (CR)	Fraction of episodes ending in obstacle contact.
Near-Miss Rate (NMR)	Fraction of timesteps where the agent occupies a cell adjacent to an obstacle. Captures latent risk that CR alone misses; a policy with 0% CR but 60% NMR is fragile.
Wait-Action Frequency (WAF)	Per-episode share of Wait actions. Detects degenerate "camping" strategies. Reviewer 3's specific ask.
Per-Room Success Rate	SR broken down by target room. Uneven difficulty across targets reveals whether the agent has learned spatial structure or memorised a single route.
Train-Test Generalisation Gap	$\text{IQM}_{\text{train}}(\text{unseen seeds}) - \text{IQM}_{\text{test}}(\text{unseen layouts})$. Measures overfitting to layout topology.
Sample Efficiency (SE)	Environment steps to reach 90% of asymptotic IQM. Reported per

	algorithm with bootstrap CI.
Empirical Lipschitz Ratio	$\text{median}[\ \pi(s) - \pi(s')\ _\infty / \ s - s'\ _1]$ divided by the analogous quantity for V , over a fixed sample of state pairs. Quantitative form of the policy-hard claim.
Wall-Clock Training Time	Per algorithm per seed. Honest reporting of cost.
Model size & inference latency	Actor parameter count, FLOPs, single-step inference time on CPU. Substantiates "resource-constrained" claims (Reviewer 1 asked for this explicitly).

4.3 Statistical reporting standard

- Every aggregate number in the paper carries a 95% CI.
- Every algorithm comparison reports an effect size (probability of improvement) in addition to a p-value.
- All raw per-seed numbers go into a public supplementary CSV alongside the code.
- All figures use seeded random colour palettes and consistent axis ranges across panels.

5. Environment Upgrades

5.1 Procedural layout generation

The single fixed map is the most damaging weakness in the original paper. We replace it with a procedural layout generator.

- **Algorithm:** Binary space partitioning (BSP) over a 20×20 grid. Generate 4 rectangular rooms, connect via a corridor or doorway between adjacent partitions, place 4–8 obstacles per room.
- **Seed control:** Layout RNG seed is separate from training seed, so layouts are reproducible.
- **Train/test split:** Generate 200 layouts; reserve 50 as a held-out test set, never seen during training.
- **Difficulty stratification:** Tag each layout with a difficulty score (A*-shortest-path length from hallway centre to each room). Report test results stratified by difficulty quartile so easy-test bias is exposed.
- **Backward compatibility:** The original fixed 20×20 layout is included as a single test instance, so legacy results remain comparable.

5.2 Fixing the POMDP framing

Reviewer 1 was correct: providing normalised global (x, y) makes the task closer to fully observed. Two paths forward; we'll run both and report.

- **Variant A — strict POMDP:** State = [proximity (5), target one-hot (4), relative target vector (2), Δ steps remaining (1), local region type (3)]. No global coordinates. 15-d.
- **Variant B — original-style (rebrand as MDP):** Keep the 16-d state but call it what it is: a low-dimensional MDP with engineered features. Drop POMDP claims.
- **Variant C — POMDP + LSTM:** Use Variant A but add an LSTM encoder to the policy and critic, validating the partial-observability framing with recurrence. Reviewer 1's explicit suggestion.

Primary results table reports both A and B side-by-side. C runs once on the best inverted-asymmetry configuration as a recurrence ablation.

5.3 Dynamic-obstacle variant (Phase 3 ablation)

To address "static environment" concerns without contaminating the main asymmetry experiment.

- **Setup:** 0, 1, or 2 NPC agents move on fixed patrol cycles. Agent observes their positions in adjacent cells via the existing proximity sensors.
- **Used only as ablation:** Trained from scratch on dynamic, then transferred from static-trained checkpoints to dynamic, to separate "can learn it" from "transfers to it."
- **Restricted to the best inverted-asymmetry config:** Avoids exploding the experimental matrix.

5.4 Sensor-noise ablation

- **Noise model:** Each proximity bit independently flipped with probability $p \in \{0, 0.05, 0.10, 0.20\}$.

- **Purpose:** Reviewer 3 asked for sensor-noise ablation as a sim-to-real proxy. Cheap to run.
- **Reported as:** SR and CR degradation curves vs. p . One run per noise level on the best config.

6. Experimental Plan

The plan has four phases with explicit go/no-go gates between them. Phase 1 is gating: if the asymmetry result does not survive multi-seed replication, the narrative shifts and Phase 2's emphasis changes.

6.1 Phase 1 — Multi-Seed Replication (gating)

Goal: Determine whether the inverted asymmetry result survives proper statistical reporting.

What to run: Every configuration from the original Tables 1 (DQN ×5) and 2 (PPO ×4) at 10 seeds each, plus Double DQN, Dueling DQN, and A2C baselines at the closest-matching architectures. 90 total training runs.

Compute estimate: $\approx 200\text{k env steps per run} \times 90 \text{ runs} \approx 18\text{M env steps total}$. At $\sim 5\text{k steps/sec}$ on Colab Pro CPU rollouts with PyTorch updates, this is $\sim 1000 \text{ GPU-hours}$. On a single Colab Pro account at $\sim 24 \text{ GPU-hours/day}$ of sustained use, this is ~ 6 calendar weeks of pure compute. Budget 8 weeks for disconnect overhead. With 2 people, the work is owner-split (Samuel runs configs, collaborator builds infrastructure in parallel) rather than parallelisable on the GPU itself.

Outputs: reliable-style results table with IQM $\pm 95\%$ CI for every configuration. Performance profiles. Per-action frequency tables. POR computed against A*.

Decision gate: Does Exp 4 (inverted asymmetry) significantly beat Exp 1 (symmetric) and Exp 3 (deep-critic) at 95% CI?

- If YES $\rightarrow$ proceed to Phase 2 with asymmetry as the main claim.
- If NO but H1 partial (beats one of the two) $\rightarrow$ proceed to Phase 2 but with the broader "architectural ablations" framing.
- If NO entirely $\rightarrow$ the paper becomes a methodological one: "single-seed RL claims don't replicate." Still publishable, different narrative.

6.2 Phase 2 — Capacity-Ratio Ablation + Generalisation

Goal: Isolate the actor:critic ratio as the independent variable; introduce procedural layouts; fix the POMDP framing.

What to run:

- **2A. Capacity-Ratio Sweep.** 5 total-parameter budgets $\times$ 5 actor:critic ratios = 25 configs $\times$ 5 seeds = 125 runs.
- **2B. Procedural Generalisation.** Best 3 configurations from Phase 1 trained on {10, 50, 200} layouts, evaluated on 50 held-out layouts. $3 \times 3 \times 5$ seeds = 45 runs.
- **2C. POMDP Variants.** Variants A, B, C on the best config $\times$ 5 seeds = 15 runs.

Total: ≈ 185 runs. Estimated 350 GPU-hours.

Outputs:

- Heatmap: total-params (y) $\times$ ratio (x), colour = IQM. Reviewer-3's central question is answered visually.
- Generalisation gap curves: train-IQM vs. test-IQM as N_{train} grows.
- POMDP variant comparison table.

6.3 Phase 3 — Cross-Environment Validation + Lipschitz + Dynamic Ablation

Goal: Test whether the hypothesis transfers beyond our custom env; replace the qualitative Lipschitz argument with measurement; address static-only criticism.

- **3A. MiniGrid Validation.** Run symmetric vs. inverted PPO on MiniGrid-FourRooms, MiniGrid-MultiRoom-N2-S4, and MiniGrid-DoorKey-6x6. 3 envs $\times$ 2 configs $\times$ 5 seeds = 30 runs. Owned by Samuel (training queue) and collaborator (env integration).
- **3B. Lipschitz Measurement.** Post-hoc on saved Phase 1+2 checkpoints. Sample 10k state pairs, compute and plot $\text{Lip}(\pi)$ and $\text{Lip}(V)$ distributions across architectures. No new training.
- **3C. Dynamic-Obstacle Ablation.** Best inverted config trained-from-scratch and transferred-from-static on dynamic env. 2 modes $\times$ 5 seeds = 10 runs.
- **3D. Sensor-Noise Ablation.** Best config $\times$ 4 noise levels $\times$ 3 seeds = 12 runs.

Total: $\approx$ 52 runs + $\sim$ 30 GPU-hours of Lipschitz analysis. Estimated 280 GPU-hours (MiniGrid is the largest single line item — $\sim$ 5–7M env steps per run with CNN encoder).

6.4 Phase 4 — Hybrid Safety Architecture + Optional Stretch

Goal: Implement the proposed PPO+DQN hybrid; produce final figures; optional GRPO comparison if time allows.

- **4A. Hybrid Implementation.** PPO planner outputs an action distribution; DQN safety filter masks any action whose predicted Q value falls below a learned safety threshold (or whose next-state proximity sensor activates). Evaluate on test layouts.
- **4B. Figure regeneration.** All figures redone in seaborn with consistent style. Replace Figures 2–5 with reliable-style learning curves + performance profiles.
- **4C. (Optional) GRPO.** Implement GRPO with group-normalised advantages over a small batch of trajectories. Compare against PPO on the best configuration. Only if Phases 1–3 finish on schedule.
- **4D. Writing.** Full paper rewrite with new results.

7. Cross-Environment Validation (Phase 3A) — Detail

The reviewers did not ask for this explicitly, but their generalisation critique implies it. Validating inverted asymmetry on standard sparse-reward benchmarks is what converts this from a single-domain curiosity to a defensible architectural finding. Keeping this in scope is the main reason the timeline was extended from 16 to 20 weeks.

7.1 Selected environments

Environment	Why it tests the hypothesis	Standard protocol
MiniGrid-FourRooms-v0	Topologically similar to our env: 4 rooms + corridors + doorways. Sparse reward, partial observability via 7×7 view.	PPO with shared CNN encoder. Standard hyperparameters from MiniGrid baselines. 5M env steps.
MiniGrid-MultiRoom-N2-S4-v0	Longer-horizon corridor navigation. Sparse reward. Tests whether asymmetry helps in extended sequences.	Same as above; 7M env steps.
MiniGrid-DoorKey-6x6-v0	Combinatorial sparse reward (pick up key → open door → reach goal). Tests whether asymmetry generalises to sub-goal structure.	Same as above; 5M env steps.

7.2 What success looks like

- Inverted asymmetry achieves higher IQM than symmetric at $p < 0.05$ on ≥ 2 of 3 environments → H5 supported, hypothesis generalises.
- Symmetric ties or wins → H5 falsified; the result is domain-specific. Still publishable, framed as "works in residential-style layouts but does not transfer."

7.3 Why MiniGrid specifically

MiniGrid is the de facto standard for sparse-reward partially-observable navigation benchmarking, with PPO and A2C baselines well-tuned in the literature and small enough environments that 5–7M env steps fit Colab Pro budgets. Procgen, an alternative, requires CNN encoders and 25M+ env steps per run; that's out of compute budget.

7.4 Compute realism (2-person, 1 Colab Pro)

MiniGrid is the most expensive Phase 3 line item: ~5–7M env steps × 30 runs × 30% of our usual throughput (CNN slows things down). Estimated ~200 GPU-hours of dedicated Phase 3A time, ≈ 10 calendar days. Mitigated by: (1) only 2 envs in worst case (drop MultiRoom if time-constrained), (2) 3 seeds instead of 5 as a fallback (still gives bootstrap CIs, just wider), (3) MiniGrid jobs can overlap with Phase 2 writing and Phase 3B/C/D analysis.

8. Optional Stretch Goals

These are valuable but not gating. Only attempt after Phases 1–3 are complete and on schedule. Better to ship a tight paper than a sprawling one.

8.1 GRPO comparison

- **What:** Group Relative Policy Optimization replaces the value baseline with a group-normalised advantage computed over a batch of trajectories from the same prompt/state.
- **Why interesting:** Directly addresses the asymmetry question by removing the critic entirely. If GRPO matches or beats inverted-asymmetry PPO, that's a strong story.
- **Why risky:** GRPO is designed for LLM RL with verifiable binary rewards. Adapting it to dense-reward grid navigation requires choices that reviewers could attack.
- **Effort:** Reimplementation $\approx$ 1 week; experiments $\approx$ 1 week of GPU time. Only do this if Phases 1–3 finish on time.

8.2 Behavioural cloning safety filter alternative

- **What:** A simple rule-based safety filter ("never take an action whose next-state proximity sensor is 1") as a non-RL alternative to the DQN safety layer.
- **Why:** Reviewer 3 suggested this. Strengthens the hybrid analysis by showing whether the DQN-as-safety-layer is necessary or whether a 5-line rule does the same job.
- **Effort:** Trivial implementation; $\sim$ 1 day of experiments.

8.3 Sim-to-real gap analysis (writeup only)

- **What:** Quantitative discussion of what features of the sim env are most likely to break under physical conditions, citing recent sim-to-real RL literature.
- **Why:** Cheap way to address Reviewer 3's deployment concerns without actual hardware.
- **Effort:** Half a writing day.

9. Team Allocation

Two people. With a single Colab Pro account, training is serial — one configuration at a time on the GPU. The split below puts Samuel on the experiments-and-claims path and the collaborator on the infrastructure-and-analysis path, so the two streams run in parallel without contending for the GPU.

Role	Who	Responsibility
Lead author	Samuel	Owens the asymmetry hypothesis. Designs and runs Phase 1 training configurations and Phase 2 capacity sweep. Maintains the reliable analysis pipeline. Drives the paper rewrite. Final call on narrative shifts at the Phase 1 gate.
Co-author	Collaborator	Owens the Environment subsystem: procedural layout generator, POMDP variants, dynamic-obstacle wrapper, sensor-noise wrapper, and MiniGrid integration (CNN encoder, observation wrappers, hyperparameter porting). Runs Phase 2 procedural-generalisation experiments and assists Samuel on Phase 3A MiniGrid runs. Implements and runs Phase 3 Lipschitz measurement, dynamic-obstacle ablation, sensor-noise ablation. Implements the Phase 4 hybrid safety filter. Generates final figures. Writes the methods, environments, and ablation sections of the paper.

The two streams are designed not to block each other. Samuel can run training even when the collaborator is mid-development on a new wrapper, because the wrappers live behind clean Env interfaces and the trainer doesn't depend on the next wrapper landing. Conversely, the collaborator can develop and test new wrappers on the existing env while Samuel runs the long Phase 1 jobs.

9.1 Sync cadence

- **Twice-weekly:** 30-minute syncs (Monday: plan the week's runs; Friday: review what landed). Lower coordination cost than weekly hour-long meetings; fits 2-person teams better.
- **Async first:** Default to W&B run-board updates and PR descriptions for status, not meetings. Synchronous time is reserved for decisions, not status.
- **Phase gates:** 90-minute review meeting at the end of Phase 1 (gating), Phase 2, and Phase 3. Phase 1 gate decides whether asymmetry survives as the headline claim.
- **Shared infra:** Single Weights & Biases project (arch-ablations-sparse-reward). One CSV-of-truth per phase, version-controlled. One shared LaTeX repo (Overleaf or local) for writing. Single GPU queue: write to it in a #gpu-queue channel before kicking off a run.

9.2 Authorship discussion (do early)

With two people doing substantially different work, authorship is simple: Samuel is lead author; collaborator is second author. Have the conversation explicitly at the kick-off meeting anyway, and write down CRediT contributions as work happens (don't reconstruct them at submission). The first-vs-second-author distinction matters for promotion and grant applications; address it openly now rather than after the paper is written.

10. Compute Budget

Estimates assume PPO at ~3k env steps per second of training (including gradient updates) on a single T4 GPU, DQN at ~5k env steps per second, and MiniGrid runs at ~2k env steps per second (CNN encoder).

Phase	Runs	Steps / run	GPU-hours
Phase 1 (multi-seed replication)	90	200k	≈ 1000
Phase 2 (capacity ratio + generalisation + POMDP)	185	200k	≈ 350
Phase 3A (MiniGrid)	30	5M avg	≈ 200
Phase 3B (Lipschitz, post-hoc)	—	—	≈ 30
Phase 3C/D (dynamic + noise)	22	200k	≈ 80
Phase 4 (hybrid + viz)	10	200k	≈ 30
Total	≈ 340	—	≈ 1690

10.1 Reality check

- Colab Pro = ~24 GPU-hours/day at sustained use per account. 1 account → ~170 GPU-hours/week. 1690 total ÷ 170/week ≈ 10 weeks of pure compute IF the account runs uninterrupted. It will not. Budget 12–14 weeks across Phases 1, 2, and 3 combined.
- Run management matters: failed Colab sessions waste hours. Use checkpointing every 25k env steps. Resume on disconnection. Pre-commit a resume-from-checkpoint script in Week 1.
- Reduce-budget fallbacks if Phase 1 overruns:** (a) drop seeds from 10 to 7 in Phase 1 (saves ~300 GPU-hours / ~2 weeks); (b) drop MultiRoom from MiniGrid in Phase 3A (saves ~70 GPU-hours); (c) defer GRPO to a follow-up paper (saves ~30 GPU-hours).
- The 20-week timeline is the binding constraint.** If at the Phase 1 gate (week 11) Phase 1 isn't finished, apply fallback (a) immediately rather than letting the project slip further.

11. Timeline

20-week plan, assuming Phase 1 starts after env code is finalised. The extension from an earlier 16-week draft buys back time for cross-environment validation and accounts for serial (single-GPU) training. Owners are listed by stream (S = Samuel, C = Collaborator) — both people are usually engaged each week but the listed owner is the deliverable's accountable lead.

Week	Phase	Deliverables	Owner
1	Prep	Env code finalised; procedural generator scaffolded; rliable + W&B project set up; CLAUDE.md committed; reading: Agarwal et al. 2021.	S + C
2–3	Phase 1 setup	Training scripts converted to multi-seed; baselines (Double DQN, Dueling DQN, A2C) implemented; A* shortest-path for the fixed map; first sanity runs at 3 seeds; checkpoint/resume audited.	S
4–10	Phase 1 runs	90 training runs over ~7 weeks at single-GPU throughput. Weekly W&B dashboard updates. Collaborator works on procedural generator and POMDP variants in parallel (no GPU contention).	S (runs), C (env)
11	Phase 1 gate	Narrative decision meeting. IQM table + performance profiles. Decide: does asymmetry survive? Lock title and abstract framing accordingly.	S + C
11–14	Phase 2 runs	Capacity-ratio sweep (25 configs × 5 seeds) + procedural generalisation + POMDP variants. Mid-phase results review at week 13.	S (runs), C (procedural)
13–15	Phase 3A prep	MiniGrid integration: CNN encoder, observation wrapper, hyperparameter porting from MiniGrid-baselines. Smoke-test with 1 seed on FourRooms. Parallel with Phase 2 end.	C
15–17	Phase 3 runs	Phase 3A MiniGrid runs (30 runs across 3 envs). Phase 3B Lipschitz measurement on Phase 1+2 checkpoints. Phase 3C dynamic-obstacle ablation. Phase 3D sensor-noise ablation.	S (MiniGrid), C (Lipschitz + ablations)
17–18	Phase 4	Hybrid safety architecture implementation and evaluation on test layouts. All figures regenerated in seaborn. Stretch: GRPO comparison if time.	C (hybrid), S (figures)
18–19	Writing	Full paper draft. Sections drafted as their phases finished get integrated. Internal review (both people review the whole).	S + C

20	Polish + submit	Reproducibility README. Public W&B project. Slack week for surprises (use it; do not skip).	S
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11.1 Critical-path notes

- Phase 1 (weeks 4–10, 7 weeks of GPU time) is the longest single block. Serial single-GPU training is the binding constraint. Hold a hard mid-phase checkpoint at week 7 — if not at least 50% of runs are complete by then, invoke the seed-reduction fallback.
- Phase 2 and Phase 3A prep run in parallel (week 13). They don't share the GPU — Phase 2 is on our env (cheap), Phase 3A prep is engineering work (no training yet).
- Writing starts at week 18, not week 19. Each phase's section gets drafted right after that phase ends, while results are fresh.
- Weeks 19 and 20 are protected for finishing and slack. Resist the urge to start a "quick" new experiment here. The history of academic projects is paved with quick new experiments that broke deadlines.

12. Risk Register

Risk	Likelihood	Impact	Mitigation
Inverted asymmetry does not survive multi-seed replication	Medium	High	Pivot narrative to "architectural ablations" framing. Phase 2 still produces a useful capacity-ratio result either way. The capacity sweep finding is independent of whether the original Exp 4 effect was real.
Colab Pro caps / disconnects waste compute	High	Medium	Aggressive checkpointing (every 25k steps). Resume-from-checkpoint script audited week 1. Two-account redundancy.
MiniGrid hyperparameters need substantial retuning	Medium	Medium	Use published MiniGrid PPO baselines (e.g., minigrid-baselines repo) as starting point. Don't try to beat SOTA; just compare symmetric vs. inverted at matched hyperparameters.
Collaborator dropout / illness	Medium	Very High	This is the biggest single risk on a 2-person team. Mitigation: (1) keep all infra code and configs in the shared repo, never on a single laptop; (2) Samuel cross-trains on the env subsystem during weeks 1–3 so he can pick it up; (3) if the collaborator drops out in Phase 1, paper scope drops back to single-env (the 16-week fallback plan from an earlier draft). If after Phase 2, the existing results may still support a publishable paper without Phase 3A.
Single-GPU Colab account hits sustained-use cap	Medium	High	Colab Pro throttles after long sessions. Schedule training across both people's waking hours (one runs overnight in their timezone, the other during their day). If throttling becomes blocking, fall back: reduce Phase 1 seeds to 7. Budget for ~\$50/month on Colab Pro+ as a contingency.
Hybrid safety architecture turns out unstable	Medium	Low	Phase 4 deliverable. If it fails, the paper still has Phases 1–3 results. Phase 4A becomes "future work, attempted and reported" which is itself publishable.
Reviewers ask for things we didn't plan for	High	Low	Build a 1-week buffer between submission and any deadline. Have stock answers prepared for common asks: "why not Rainbow?", "why not procgen?", "why not real hardware?".
Statistical analysis errors in reliable usage	Low	High	Pair-review every analysis notebook before its results enter the paper. Use the reliable colab as a template.

Layout generator produces unsolvable maps	Medium	Medium	After generating each map, run A* from start cell to target cells; reject any map where any target room is unreachable. Filter at generation time.
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13. Venue Decision Tree

Venue choice depends on what survives Phase 1 and Phase 2. Final decision at the end of Phase 3.

13.1 If all hypotheses survive (best case)

- **Primary target:** ICML or NeurIPS workshops in the year-end cycle (RL Workshop, Robot Learning Workshop, Safe and Robust Control Workshop). These accept high-quality empirical findings with clear single claims; perfect for a multi-environment ablation paper.
- **Reach target:** AAAI / IJCAI main track. Possible if Phase 3 shows clean cross-env generalisation.

13.2 If H1 survives but H3 / H5 (generalisation, cross-env) are weaker

- **Primary target:** Domain venue (IROS, ICRA — robotics; ASSETS — assistive tech). These accept work that is strong in one domain even without strong cross-domain claims.
- **Alternative:** RLC (Reinforcement Learning Conference) — accepts careful empirical papers, less obsessed with breadth.

13.3 If H1 fails entirely

- **Primary target:** Workshop on Reproducibility in ML (NeurIPS) or similar. "What we believed from a single seed didn't replicate, and here is what did." This is genuinely valuable and increasingly publishable.
- **Alternative:** Lessons-learned writeup as a technical report on arXiv. Not all good projects yield top-venue papers; an honest negative result is still a contribution.

13.4 Submission hygiene regardless of venue

- Public, de-anonymised GitHub repo with reproducibility instructions and per-seed CSVs.
- Hyperparameter sweep configurations saved as YAML, not buried in code.
- All figures regenerable from a single `make figures` command.
- CRediT contribution statements written up before authorship discussion.

14. Appendices

Appendix A — Code structure changes

The current repo (Environment/, Training/, dqn_models/, ppo_models/, main.py) needs the following additions before Phase 1 starts.

- Environment/procedural.py — BSP-based layout generator, layout difficulty scoring, train/test split.
- Environment/dynamic.py — moving NPC wrapper; subclasses ResidentialGridEnv.
- Environment/noise.py — sensor-noise wrapper.
- Training/seeds.py — seed manager: ensures every run logs (algorithm, config_id, seed) tuple uniquely.
- Training/baselines/ — Double DQN, Dueling DQN, A2C implementations (or stable-baselines3 imports with matched hyperparameters).
- Analysis/rliable_analysis.ipynb — IQM, performance profiles, probability of improvement; single source of truth for all paper figures.
- Analysis/lipschitz.py — empirical local Lipschitz estimator using saved checkpoints.
- Analysis/astar.py — shortest-path computation for Path Optimality Ratio.
- Analysis/trajectories.py — qualitative behaviour viz: heatmaps of agent occupancy, sample trajectory overlays.
- hybrid/safety_filter.py — PPO+DQN hybrid architecture (Phase 4).

Appendix B — Reading list (cited or planned)

- **Statistical reporting:** Agarwal, Schwarzer, Castro, Courville, Bellemare. "Deep Reinforcement Learning at the Edge of the Statistical Precipice." NeurIPS 2021. (rliable.)
- **Asymmetric actor-critic:** Liu et al. "Asymmetric Proximal Policy Optimization: Mini-Critics Boost LLM Reasoning." arXiv 2025. (Already cited.)
- **Informed asymmetric AC:** Ebi, Lambrechts, Ernst, Böhm. ERLW 2025. (Already cited.)
- **MiniGrid benchmark:** Chevalier-Boisvert et al. MiniGrid suite. To cite the Gymnasium-MiniGrid 2023 paper.
- **GRPO:** Shao et al. DeepSeek-Math GRPO 2024. Cite if Phase 4C runs.
- **Sparse-reward RL:** Hu et al., "Variational Information Maximization" for sparse-reward A2C in MiniGrid — for related work motivation.

Appendix C — What the original paper still gets right (keep, don't rewrite)

The rework should not throw away these parts — they remain solid and motivate the work:

- The assistive-tech motivation in Section 1. Stays as-is.
- The literature review on indoor navigation technologies (§2.1). Light cleanup only.
- The actor-critic asymmetry framing (§2.4). This is the contribution; refine but keep.
- The reward design (§3.5). Engineered with care; defensible.
- The capacity-mismatch hypothesis as motivation. The hypothesis is fine; the evidence was weak.

Appendix D — Glossary of terms used

- IQM — Interquartile Mean. Mean of the middle 50% of scores after dropping bottom and top 25%. Robust to outliers; standard RL aggregate metric since Agarwal et al. 2021.
- Stratified bootstrap CI — resample seeds with replacement $B=2000$ times; the 2.5–97.5 percentiles of the bootstrap distribution give the 95% CI. Makes no distributional assumptions.
- `rliable` — Google Research's Python library implementing the above. Drop-in for the project.
- POR — Path Optimality Ratio. $\text{agent_steps} / A^*_optimal_steps$. 1.0 is optimal.
- $\text{Lip}(f)$ — Lipschitz constant of function f . Quantifies how rapidly f changes across input space.
- Capacity ratio — $\text{actor_params} / \text{critic_params}$. > 1.0 means deeper actor (inverted asymmetry).
- Generalisation gap — train-performance minus test-performance on held-out layouts.